2019 EXAMINATIONS

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.204 Software Design

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Provide the definitions of a functional requirement (FR) and a non-functional requirement (NFR). Give three examples of **NFR**s for a notebook computer.

[5 marks]

1.b The following is a part of concept description for a conference refereeing system. Draw a Statechart diagram of the system.

Conference Refereeing System (CRS)
Professional conferences are held in order to announce and discuss new results. The core activity of organizing a conference centres on selecting the papers to be presented. Usually this is done by making an open invitation calling for papers to be submitted, circulating the submitted papers to a (geographically distributed) panel of reviewers to make an assessment of suitability and quality, then selecting the best papers to appear on the programme. A system to automate the conference refereeing process should enable the following: 1. Authors submit papers to the system and a program chair (PC) of the conference checks the validity of submitted papers. 2. After a deadline, the PC uses the system to distribute the submitted papers among reviewers. Each paper is sent to three distinct reviewers, none of whom is an author of the paper. 3. The reviewers submit their reviews to the system before an agreed deadline. 4. The PC makes decisions on which papers to accept/reject. 5. The PC notifies the authors about the review results. 6. Authors of accepted papers submit revised final version papers. The PC also checks the submissions are correct. 7. The PC can cancel the conference at any time and this will initialise the CRS.

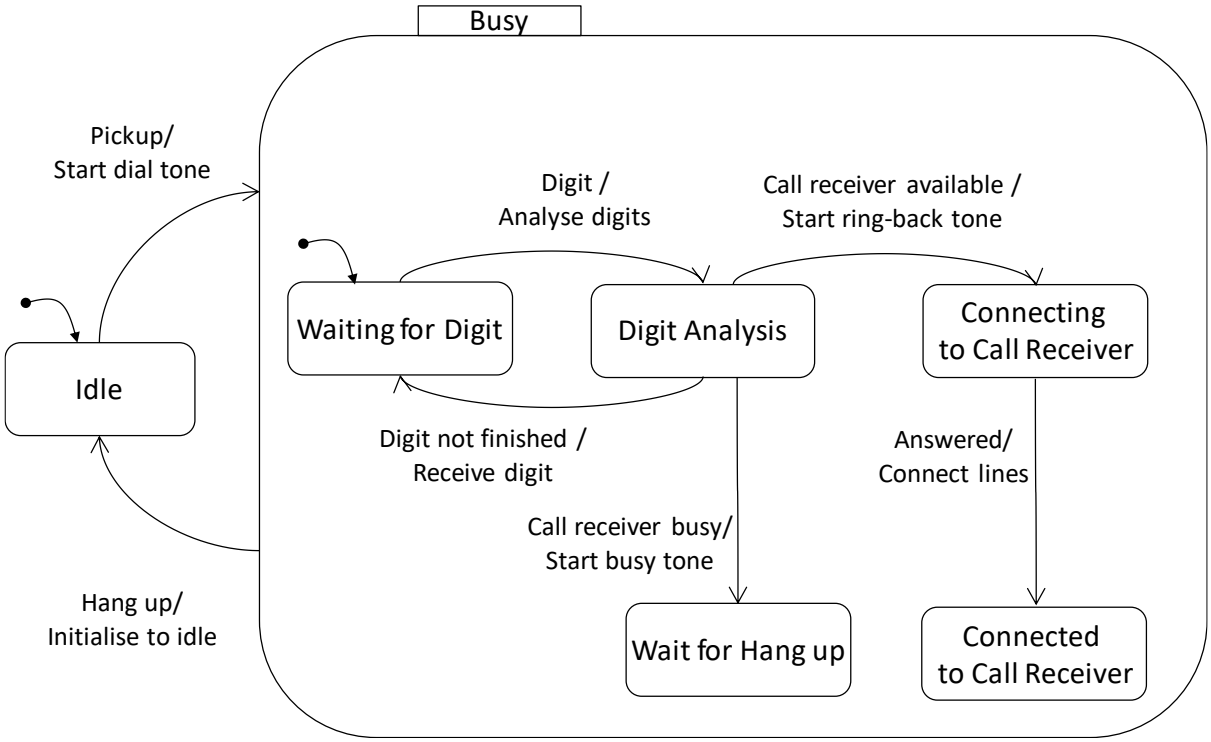
[10 marks]

Question 1 continues on the next page…

Question 1 continued.

1.c The State design pattern can be used to define states in a Statechart diagram. Draw a class diagram, which uses the State pattern to define the states of the following Statechart that captures the behaviour of a caller using a landline telephone.

[10 marks]



Total 25 marks

Question 2

2.a Draw a class diagram of the Observer design pattern. The class diagram should include participating classes, their relations with multiplicity and required methods for the design pattern.

[7 marks]

2.b Consider the following program and answer the questions below.

[11 marks]

```
public class Main {
    public static void main(String[] args) {
        NewPrint p = new MyPrint("Lancaster");
        p.printType1();
        p.printType2();
    }
}

public interface NewPrint {
    public abstract void printType1();
    public abstract void printType2();
}

public class OldPrint {
    private String string;
    public OldPrint(String string) {
        this.string = string;
    }
    public void showWithPlus() {
        System.out.println("+++" + string + "+++");
    }
    public void showWithStar() {
        System.out.println("***" + string + "***");
    }
}

public class MyPrint extends OldPrint implements NewPrint {
    public MyPrint(String string) {
        super(string);
    }
    public void printType1() {
        showWithPlus();
    }
    public void printType2() {
        showWithStar();
    }
}
```

i. Draw a class diagram that is equivalent to the program. The class diagram should include participating classes, their relations and required methods.

(7 marks)

ii. Identify the design pattern used in the program and explain when it is used.

(2 marks)

iii. Write down the result of executing the Main class.

(2 marks)

Question 2 continues on the next page…

Question 2 continued.

2.c Consider a Smart Home System (SHS) that consists of smart devices in a house and communicates with a GP and emergency services. The SHS Controller manages the following three smart devices:

1) Smart Fridge (SF): The SF is able to read, store, and communicate Radio Frequency Identification (RFID) information on food packages. The SF detects the presence of spoiled food and receives a diet plan to be monitored based on what food items the Smart Home user is consuming. When an abnormal event happens (i.e., spoiled food event or diet plan violation event), the SF reports the event to the SHS Controller.
2) Smart Cup (SC): An important part of the user's health is to ensure minimum liquid intake and the SC monitors the user's liquid consumption. When the liquid intake consistently falls below a minimum threshold, the SC notifies the SHS Controller of this event (i.e., low liquid intake event).
3) Climate Manager (CM): The CM is equipped with 2 temperature and 4 humidity sensors and controls a boiler and humidifiers to maintain desired temperature and humidity levels set by the user. When unexpected changes in temperature and/or humidity levels are detected and they cannot be resolved within five hours, the CM notifies the SHS Controller of this event (i.e., abnormal climate change event).

When the spoiled food event or the diet plan violation event is received from the SF, the SHS Controller reports this to the user's GP (medical doctor). When the low liquid intake event is received from SC for three consecutive days, the SHS Controller contacts the emergency service. When the abnormal climate change event is received from the CM, the SHS Controller contacts the emergency services.

Draw a use case diagram of the SHS Controller.

[7 marks]

Total 25 marks

Question 3

3.a Describe any 4 key objectives of software design.

[4 marks]

3.b A bank owns a number of branches countrywide. Each branch is responsible for a number of Automated Teller Machines (ATMs) in its locality. Each automated teller machine (ATM) contains an embedded software system to drive the machine hardware and to communicate with the bank's customer account database. The system accepts customer requests and dispenses cash, displays balance, displays and prints a short statement, allows PIN change. All ATM users are issued with ATM card and a personal identification number (PIN) that they must use to access the ATM services. The ATM validates the cards locally, but communicates with a remote customer database to validate PINs, verify transactions and update accounts.

Use a box and line diagram to design an architecture for the ATM system based on an appropriate architectural style. Your design should identify the ATM system's main sub-systems and the data flows between them. Be sure to state your choice of architectural style.

[5 marks]

3.c A company uses an object-oriented software system to manage its paper recycling business. The system comprises of an item manager that keeps a record of the paper being recycled and controls a paper sorter and recycling processors. The paper sorter separates the different types of paper then sends them to the recycling processors. Each type of paper has its own recycling processor.

The company recycles two types of paper: office paper and cardboard. Each batch of paper has a batch number and a quantity measure that represents the total volume of paper in the batch. Quantity is measured in cubic metres (i.e. m^3). For example, a paper with batch number "7809" and quantity "700", means the paper belongs to batch "7809" whose volume is $700m^3$. The object-oriented model of the paper recycling system is shown in **Figure 1** (Figure 1 is found on the next page).

Question 3 continues on the next page…

Question 3 continued.

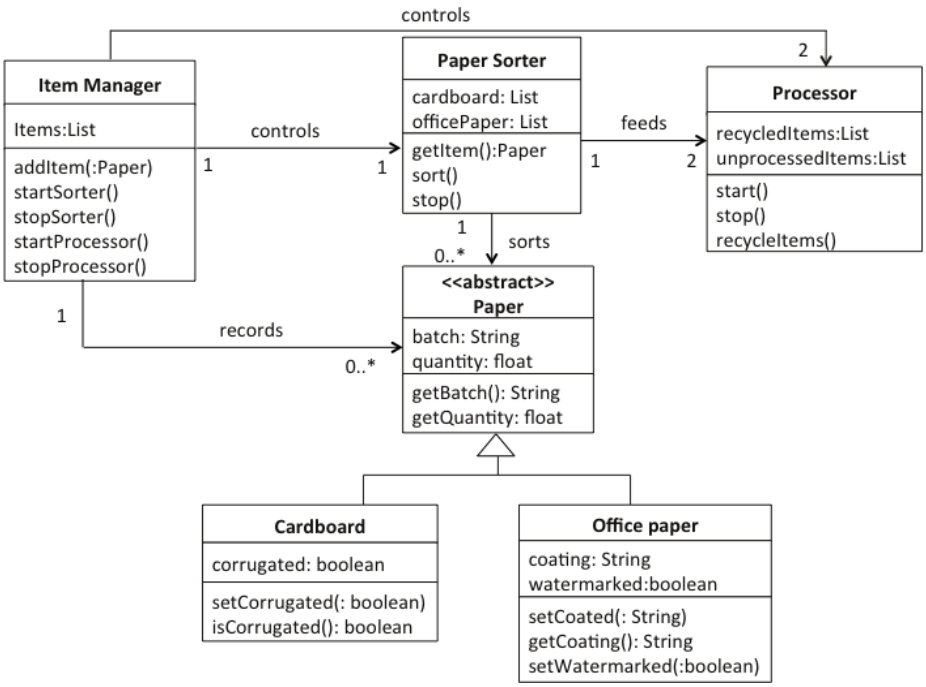


Figure 1 Recycling system model

The company wants to expand its business to recycle glass and batteries in addition to paper. In order to do this, the company will need separate recycling processors for glass and batteries. The company will continue to use batch numbers and quantity measurement to keep track of glass and batteries being recycled. However, the quantity measurement for glass and batteries will be computed in kilograms (kg). The company also wants the new system to record a description of the chemical composition for each battery and a description of glass type for each glass.

Without using abstract classes (this includes not using the abstract class 'Paper' in Figure 1), redesign the class diagram in **Figure 1** to show how the system might be modified to support the recycling of paper, glass and batteries.

[16 marks]

Total 25 marks

Question 4

4.a Explain why design conflicts might arise when designing an architecture for which both performance and reuse are the most important non-functional requirements.

[4 marks]

4.b Briefly explain 5 key differences between interface and abstract classes

[5 marks]

4.c A large consumer electronics company allows its customer to order products over the phone or via the company website. Customers are required to have an account before they can place an order. An account can be set up over the phone or via the company website. Each customer of the company has a unique identifier, address, phone number and email address. A customer using the company website (i.e. a Web user) has a username, a password and user state. User state indicates whether the web user is "active", "blocked" or "banned".

Each customer can have only one account. An account has a unique identifier, billing address, and status, which indicates whether the account is active or closed. The account also includes a shopping cart and customer orders. Each customer order has an associated order number, date that the order was placed, shipping address for the order and order status. Order status indicates if the order is "pending", "cancelled", "has been shipped" or "has been delivered". A shopping cart has the date it was created and the list of products it contains. Each order may refer to zero or more payments. Each payment has a unique identifier and is related to the customer account.

A product may have one or more line items associated with it. For example, a television brand name may have several models associated it. Each line item represents one product.

Use the UML notation to develop a class diagram for the shopping system showing the classes that make up the system, their attributes and operations. Indicate the relationships and multiplicities between the classes.

[16 marks]

Total 25 marks

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